

Lec: Customer Segmentation

Agenda:-

① Perform customer segmentation using
RFM (wrt SQL)

Problem Start:

The retail chain sells many items and needs to optimize marketing spend.

Relying on a single metric (like purchase frequency or revenue) is flawed because it doesn't capture overall customer value or behavior.

They need a way to identify which customers are most important for the business.

Excel

SQL / pandas

① Data is relatively
small

② Big Data

The RFM Segmentation can be executed using these five steps:

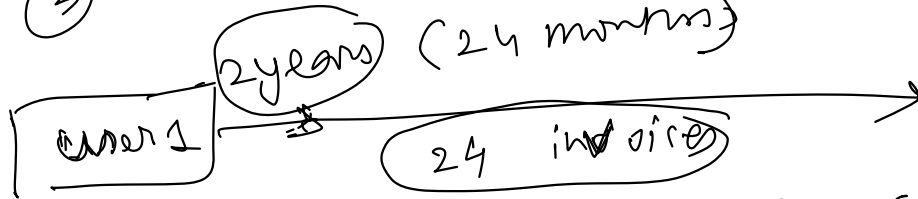
- ✓ 1. Data processing
- ✓ 2. Compute for recency, frequency, and monetary values per customer
- ✓ 3. Determine quantiles for each RFM metric
- ✓ 4. Assign scores for each RFM metric
- ~~5. Define the RFM segments using the scores in step 4~~

RFM scores per customer

① Monetary: Sum of sales

② Frequency: $\frac{\# \text{ of invoices}}{\# \text{ of months}}$

③ Recency: Last purchase date.



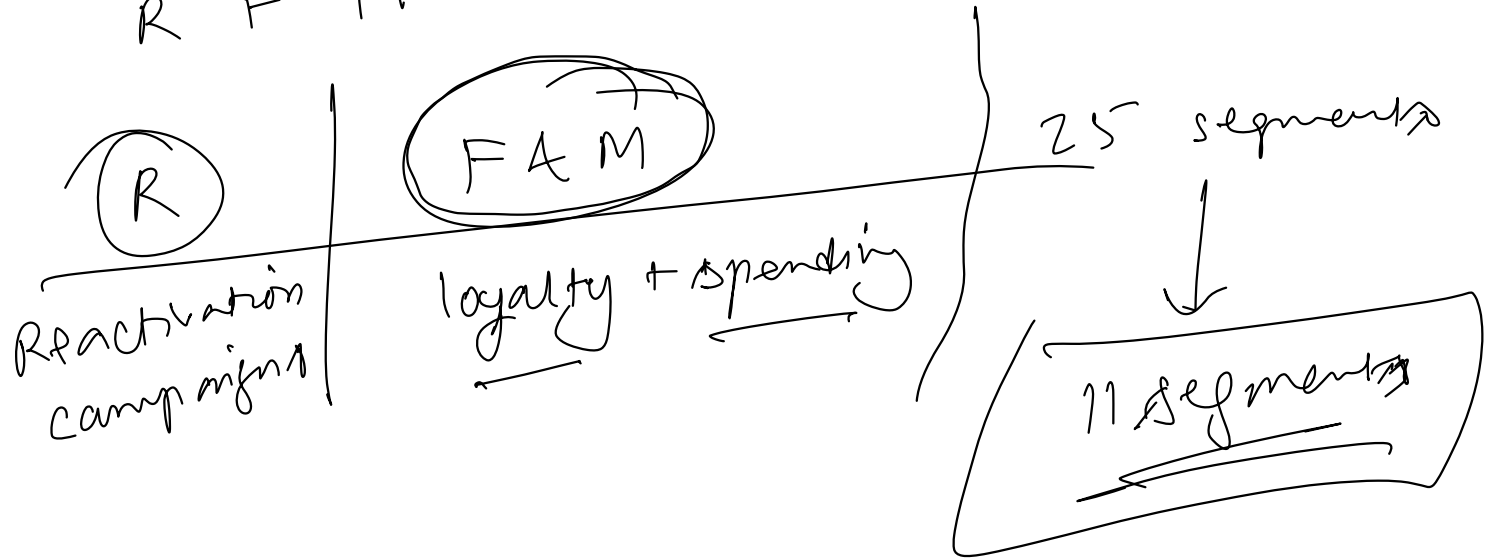
Frequency,
Rate 1 invoice/
month.



Frequency
Rate
2 invoices/month.

$3 \times 3 \times 3 = 27$ segments

$5 \times 5 \times 5 = 125$ segments
R F M



11 segments

Customer Segment	Activity	Actionable Tip
Champions	Bought recently, buy often and spend the most!	Reward them. Can be early adopters for new products. Will promote your brand.
Loyal Customers	Spend good money with us often. Responsive to promotions.	Upsell higher value products. Ask for reviews. Engage them.
Potential Loyalist	Recent customers, but spent a good amount and bought more than once.	Offer membership / loyalty program, recommend other products.
Recent Customers	Bought most recently, but not often.	Provide on-boarding support, give them early success, start building relationship.
Promising	Recent shoppers, but haven't spent much.	Create brand awareness, offer free trials
Customers Needing Attention	Above average recency, frequency and monetary values. May not have bought very recently though.	Make limited time offers, Recommend based on past purchases. Reactivate them.
About To Sleep	Below average recency, frequency and monetary values. Will lose them if not reactivated.	Share valuable resources, recommend popular products / renewals at discount, reconnect with them.
At Risk	Spent big money and purchased often. But long time ago. Need to bring them back!	Send personalized emails to reconnect, offer renewals, provide helpful resources.
Can't Lose Them	Made biggest purchases, and often. But haven't returned for a long time.	Win them back via renewals or newer products, don't lose them to competition, talk to them.
Hibernating	Last purchase was long back, low spenders and low number of orders.	Offer other relevant products and special discounts. Recreate brand value.
Lost	Lowest recency, frequency and monetary scores.	Revive interest with reach out campaign, ignore otherwise.

Other segmentation rules:

✓ ① Technographic segmentation

✓ ② Customer behaviour "

✓ ③ Needs-based "

✓ ④ Customer status "

↳ RFM

} Customer
segmentation
models

